

GROUNDWATER PUMPS

Comprehensive Engineering Module

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1 8.1 Classification of Pumps

1.1 Introduction

Groundwater pumps are mechanical devices designed to lift water from a lower potential energy level (the aquifer/well) to a higher potential energy level (ground surface or overhead tank). In hydraulic engineering, the pump adds energy to the fluid to overcome gravity, friction losses in pipes, and velocity head requirements.

1.2 General Classification

Pumps used in groundwater engineering are broadly categorized based on the mechanical principle used to move the water.

1.2.1 1. Variable Displacement Pumps (Rotodynamic)

In these pumps, kinetic energy is imparted to the fluid by a rapidly rotating element (impeller). This velocity energy is subsequently converted into pressure energy (head) as the fluid exits the pump casing. The discharge varies inversely with the head (i.e., as lift increases, flow decreases).

- **Examples:** Centrifugal pumps, Vertical turbine pumps, Submersible pumps, Jet pumps.
- **Characteristics:** Smooth, non-pulsating flow; high discharge capacity; simple construction.

1.2.2 2. Positive Displacement Pumps

These pumps operate by trapping a fixed amount of water in a chamber and mechanically forcing it out. The discharge is directly proportional to the speed of the pump and is relatively independent of the head.

- **Examples:** Reciprocating (Plunger/Piston) pumps, Rotary pumps, Hand pumps.
- **Characteristics:** Pulsating flow; capable of very high heads; suitable for low discharge.

1.2.3 3. Buoyancy / Air-Lift Pumps

These pumps rely on density differences rather than mechanical parts moving the water directly. Compressed air is injected into the water column, reducing its specific gravity and causing it to rise.

- **Example:** Air-lift pump.
- **Characteristics:** Low efficiency; no moving parts down the well; excellent for sandy water.

Past Exam Questions

Past Exam Questions:

- **2073 Bhadra (Regular) [4 Marks]:** Describe in short any four common groundwater pumps.
- **2071 Bhadra [4 Marks]:** What are the types of pumps used for lifting water?

2 8.2 Reciprocating (Plunger) & Jet Pumps

While less common in large-scale municipal projects due to efficiency or capacity limitations, these pumps are vital for specific domestic or rural applications where conditions (like depth or power availability) restrict the use of standard pumps.

2.1 A. Reciprocating (Plunger) Pumps

This is a classic Positive Displacement pump widely used in windmills and hand pumps.

2.1.1 Working Principle

The pump consists of a cylinder, a piston (plunger), a connecting rod, and two one-way check valves: the **Suction Valve** (at the bottom of the cylinder) and the **Discharge Valve** (on the piston or outlet).

1. **Suction Stroke:** As the piston moves upward (or backward), the volume inside the cylinder increases, creating a partial vacuum. The pressure difference causes the Suction Valve to open, drawing water from the well into the cylinder. The Discharge Valve remains closed.
2. **Discharge Stroke:** As the piston moves downward (or forward), the volume decreases, pressurizing the water. The Suction Valve is forced shut, and the Discharge Valve opens, forcing the trapped water out of the cylinder and up the delivery pipe.

2.1.2 Suitability & Characteristics

- **High Head Capability:** Since water is mechanically forced, it can be lifted to very great heights, limited only by the strength of the cylinder and the power available.
- **Efficiency:** High mechanical efficiency (80-90%) when seals are new.
- **Discharge:** Delivers a pulsating, intermittent flow. Discharge is generally low compared to centrifugal pumps.
- **Constraints:** Not suitable for sandy water. Sand grains get trapped between the piston and cylinder wall, rapidly destroying the leather or rubber seals (cups).

2.2 B. Jet Pumps

A Jet Pump is a hybrid device combining a surface **Centrifugal Pump** with a subsurface **Ejector (Jet) Assembly**.

2.2.1 Working Principle

1. **Recirculation:** The centrifugal pump at the surface pumps water. A portion of this water is diverted *back down* into the well through a separate pressure pipe.
2. **Nozzle Action:** This pressurized water passes through a converging nozzle at the bottom of the well. As per Bernoulli's principle, the high velocity through the nozzle creates a zone of low pressure (vacuum).
3. **Suction:** This vacuum sucks groundwater from the aquifer into the stream.
4. **Venturi Tube:** The mixed water (drive water + aquifer water) enters a diverging Venturi tube. Here, the velocity is converted back into pressure head, lifting the water up the suction pipe to the surface pump.

2.2.2 Suitability & Characteristics

- **Installation:** All moving parts (motor, impeller) are at the surface, making maintenance easy. Only the static jet assembly is underground.
- **Well Diameter:** The jet assembly is compact, allowing installation in small diameter wells (2 to 4 inches) where submersible pumps might not fit.
- **Efficiency:** Very low (typically 25–30%) because a large portion of energy is consumed just recirculating the drive water.
- **Application:** Popular for domestic supply in rural homes where the water table is deeper than 7m (the limit for standard suction pumps) but the cost of a submersible pump is unjustified.

3 8.3 Centrifugal Pumps

Centrifugal pumps are the most widely used mechanical devices for pumping water globally. They are Rotodynamic pumps that convert mechanical rotational energy into hydraulic pressure energy.

3.1 Working Principle

The operation relies on **Centrifugal Force** and **Angular Momentum**.

1. **Priming:** The pump casing and suction pipe must be completely filled with water before starting. If air is present, the impeller simply spins the air without creating enough vacuum to lift water (Air Binding).
2. **Rotation:** An electric motor or engine rotates the **Impeller** (a wheel with curved vanes) inside a stationary volute casing.
3. **Vacuum Creation:** As the impeller spins, water at the center (the Eye) is thrown radially outward by centrifugal force. This evacuation creates a low-pressure zone (vacuum) at the eye.
4. **Suction:** Atmospheric pressure acting on the water surface in the well pushes water up the suction pipe to fill the vacuum at the eye.
5. **Energy Conversion:** The water leaves the impeller tips at high velocity. It enters the **Volute Casing**, which has an expanding cross-sectional area. As the area increases, the velocity decreases, and kinetic energy is converted into pressure energy (Pressure Head), forcing water up the delivery pipe.

3.2 Key Components

- **Impeller:** The rotating component. Can be Closed (efficient, for clean water), Semi-Open, or Open (for trash/sandy water).
- **Casing:** The airtight chamber directing the flow.
- **Foot Valve:** A one-way check valve at the bottom of the suction pipe to prevent water from draining back into the well when the pump stops (maintains prime).

3.3 Critical Limitation: Suction Lift

This is the most important concept for Civil Engineers regarding centrifugal pumps.

- **Theoretical Limit:** At sea level, atmospheric pressure is 10.33 meters of water. Even with a perfect vacuum, atmospheric pressure can only push water up to this height.
- **Practical Limit:** Due to friction losses in the pipe, vapor pressure of water, and pump inefficiencies, the maximum practical suction lift is only **6 to 7 meters** (approx. 20-25 ft).
- **Consequence:** If the water table is deeper than 7 meters, a surface-mounted centrifugal pump **cannot** be used. The pump must be lowered (submersible) or a jet assembly must be added.

3.4 Cavitation

If the pressure at the suction side drops below the vapor pressure of water, the water boils at ambient temperature, forming vapor bubbles. When these bubbles move to the high-pressure discharge side, they collapse violently (implode). This phenomenon, known as **Cavitation**, causes severe pitting damage to the impeller and noise/vibration. To prevent this, the **Net Positive Suction Head (NPSH)** available must exceed the NPSH required by the pump.

3.5 Advantages & Disadvantages

Advantages	Disadvantages
Simple construction, low initial cost.	Limited Suction Lift (max $\approx 7\text{m}$).
Uniform, non-pulsating discharge.	Requires Priming (cannot pump air).
High efficiency (up to 85%) at design point.	Efficiency drops if head varies significantly.
Can handle suspended silt/sand better than piston pumps.	Prone to Cavitation if suction lift is too high.

Past Exam Questions

Past Exam Questions:

- **2072 Magh (Back) [6 Marks]:** Explain working principle of centrifugal pump used for lifting groundwater.
- **2072 Ashwin [4+2 Marks]:** Explain the working principle, advantages and limitations of centrifugal pump used for ground water.

4 8.4 Submersible & Deep-well Turbine Pumps

When the water table depth exceeds the 7-meter limit of centrifugal pumps, the pumping mechanism must be submerged to push the water up rather than pulling it.

4.1 A. Deep-well Vertical Turbine Pumps

These are essentially multi-stage centrifugal pumps adapted for deep wells. They were the standard before reliable waterproof motors were invented.

4.1.1 Working Principle

- **Split Design:** The **Motor** (Prime Mover) is located on the ground surface. The **Pump Bowls** (containing the impellers) are submerged deep in the well.
- **Drive Shaft:** A long, vertical steel Line Shaft runs down the center of the column pipe, connecting the surface motor to the subsurface impellers.
- **Staging:** They use multiple impellers stacked in series. Each stage adds pressure head. For a 100m deep well, 10 or 20 stages might be used.

4.1.2 Suitability

- **Power Flexibility:** Can be driven by an electric motor OR a diesel engine (via a right-angle gear drive). This is crucial in areas without electricity.
- **High Capacity:** Excellent for very high discharge and head requirements (municipal supply).
- **Limitations:** Installation is difficult and expensive due to the precise alignment required for the long shaft.

4.2 B. Submersible Pumps

These are the modern standard for tube wells, dominating the market from small domestic units to large irrigation projects.

4.2.1 Working Principle

- **Integrated Unit:** The Electric Motor and the Pump are close-coupled into a single, slender cylindrical unit.
- **Submerged Operation:** The entire assembly is lowered into the water by the discharge pipe.
- **Cooling:** The motor is sealed (waterproof) and filled with oil or water. It is positioned *below* the pump intake so that the water being sucked in flows past the motor housing, cooling it efficiently.
- **Power:** Waterproof electric cables run down from the surface control panel to the motor.

4.2.2 Comparison: Submersible vs. Vertical Turbine

Submersible Pump	Vertical Turbine Pump
Motor is underground (Silent).	Motor is above ground (Noisy).
Requires strictly Electric Power.	Can run on Diesel or Electric.
No drive shaft; easy to install.	Long drive shaft; difficult alignment.
Can work in crooked wells.	Requires perfectly straight wells (shaft bends).
Repair requires pulling the whole unit.	Motor repair is easy; pump repair is hard.

5 8.5 Air-Lift Pumps

The Air-Lift pump is a unique device that uses compressed air rather than mechanical impellers to lift water.

5.1 Working Principle

The principle is based on the reduction of specific gravity of the water column.

1. **Setup:** Two pipes are lowered into the well: a large **Eductor Pipe** (Discharge pipe) and a smaller **Air Line**.
2. **Injection:** A compressor at the surface pumps air down the air line to a nozzle/foot-piece located near the bottom of the eductor pipe.
3. **Aeration:** The injected air mixes with the water, forming a bubbly mixture (froth). This air-water mixture is significantly lighter (lower density) than the solid water surrounding the pipe in the well.
4. **Lifting:** The hydrostatic pressure of the heavy solid water outside the pipe pushes the lighter mixture up the pipe to the surface to balance equilibrium.

5.2 Submergence Ratio

For an air-lift pump to work, a significant portion of the pipe must be submerged. The efficiency depends on the **Submergence Ratio** (S_r):

$$S_r = \frac{H_s}{H_s + H_L} \times 100\%$$

Where:

- H_s = Depth of submergence (distance from water level to air nozzle).
- H_L = Lift height (distance from water level to discharge point).

Ideally, S_r should be around **60%**. If the water level drops significantly during pumping, the submergence decreases, and the pump may stop working entirely.

5.3 Advantages & Disadvantages

- **Advantages:**

- **Sand Handling:** No moving parts in the well means it is immune to abrasion. It is the **Best Pump** for developing new wells or pumping sandy water.
- **Simplicity:** Easy to install and remove.
- **Aeration:** Automatically aerates the water, helping remove Iron and Hydrogen Sulfide (H_2S) gas.
- **Crooked Wells:** Can easily fit into crooked holes where turbine pumps would jam.

- **Disadvantages:**

- **Low Efficiency:** Typically 20–40%. Compressing air is energy-intensive.
- **Deep Well Requirement:** It requires a deep water column (high submergence) to function. It cannot pump a well dry.

6 8.6 Selection of Pump Sets

Selecting the correct pump is a critical engineering decision. A mismatched pump can lead to immediate failure, excessive energy costs, or damage to the aquifer.

6.1 Factors Considered in Selection

6.1.1 1. Discharge Required (Q)

The pump must meet the demand (irrigation/domestic). However, the capacity must never exceed the **Safe Yield** of the well. Pumping beyond the yield causes excessive drawdown, potentially dewatering the pump intake or causing well collapse.

6.1.2 2. Total Dynamic Head (H)

The pump must generate enough pressure to lift water against all resistances.

$$H_{total} = H_{static} + s_w + h_f + h_v$$

- H_{static} : Elevation difference.
- s_w : Drawdown (water level drop) in the well.
- h_f : Friction losses in pipes and fittings.
- h_v : Velocity head ($V^2/2g$).

Selection: If $H_{static} > 7m$, Centrifugal pumps are ruled out. For very high heads (>100m), Multi-stage Submersible pumps are required.

6.1.3 3. Well Diameter

The pump must physically fit inside the well casing with clearance for cooling flow.

- A 10-inch submersible pump cannot fit in an 8-inch well.
- For very narrow wells (2–4 inches), Jet Pumps or Reciprocating pumps are often the only options.

6.1.4 4. Well Verticality (Straightness)

- **Vertical Turbine Pumps:** Require perfectly straight wells due to the rigid drive shaft.
- **Submersible Pumps:** Can tolerate slight deviations.
- **Air-Lift Pumps:** Can work in highly crooked or "dog-leg" wells.

6.1.5 5. Water Quality (Sand & Corrosion)

- **Sand:** If the aquifer yields sand, Reciprocating pumps will fail rapidly (seal wear). Centrifugal pumps are better, but Air-Lift pumps are best.
- **Corrosion:** Acidic or saline water requires Stainless Steel or Bronze impellers instead of standard Cast Iron.

6.1.6 6. Power Source Availability

- **Grid Electricity:** Submersible pumps are the most efficient and preferred choice.
- **No Electricity:** Vertical Turbine pumps (diesel drive) or Centrifugal pumps (belt drive) are required. Solar pumps are viable for low discharge needs.

6.1.7 7. Economics

- **Initial Cost:** Centrifugal pumps are cheap; Submersibles are moderate; Vertical Turbines are expensive.
- **Running Cost:** A high-efficiency submersible pump will save massive amounts of money on electricity over 10 years compared to a low-efficiency Jet or Air-Lift pump.

Past Exam Questions

Past Exam Questions:

- **2081 Final Assessment [4 Marks]:** What are the factors considered for selection of pumps?
- **2078 Chaitra [4 Marks]:** Explain about factors to be considered in the selection of pump sets for pumping of groundwater.
- **2073 Bhadra (Regular) [2 Marks]:** What are the factors to be considered while selecting the groundwater pump?